CS383 · SOFTWARE ENGINEERING

Midterm 461 Solutions

13/10/2024 · 1 hour · 15 marks

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01 Solution Q1: MCQ

#	Answer	Reason	Lecture
1	**a**	Development process. Product quality depends on how the product is developed.	5
2	**b**	Understandability: how easy it is to understand the process definition.	5
3	**d**	The schedule shows the tasks, the people and the milestones.	6
4	**d**	Consistency, Respect, Inclusion, Honesty.	6
5	**c**	Risks that affect the schedule or the quality.	6
6	**c**	Testing is a project activity, not a risk. The other three are example risks from the slides.	6

02 Solution Q2: ethics

SEEPP ethic example	Principle	Keyword
Approve software only if they believe that it is safe for use, meets specifications, passes appropriate tests	**PUBLIC**	safe for use: people's safety.
Ensure proper and achievable goals and objectives for any project	**PRODUCT**	goals and objectives of the project itself.
Assist colleagues in being fully aware of current standard work practices	**COLLEAGUES**	The word colleagues.
Improve their ability to produce accurate, informative, and well-written documentation	**SELF**	Improve their ability: self-improvement.

03 Solution Q3: plan-driven or agile

Concern	Answer	Reason
1- Emphasis on cost estimation before starting	**Plan Driven**	Everything is planned and estimated up front.
2- Flexibility in handling changing requirements	**Agile**	Agile is built around accepting change.
3- Use of detailed risk management	**Plan Driven**	Detailed, documented analysis in advance.
4- Incremental delivery of features	**Agile**	Delivery in small increments.
5- Long periods of planning and analysis	**Plan Driven**	A long planning phase before implementation.
6- Continuous integration and testing	**Agile**	One of the XP practices.

04 Solution Q4: the four activities

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1) Software specification
2) Software development
3) Software verification and validation
4) Software evolution
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The order carries marks. From Lecture 3 (the Process activities slide).

05 Solution Q5: FR and NFR

Functional = something the system **does** (a service). Non-functional = a constraint or quality of the system (speed, availability).

#	Functional Requirements
1	Retrieve customer's details
2	Update customer's details
3	Add new customer
4	Check availability of tools
5	Make an advance booking
6	Record an immediate hire
7	Record return of a tool

Also valid: Book out a tool, and Produce management summary. Any seven correct ones are enough.

#	Nonfunctional Requirements	Kind
1	Each page must load within 5 seconds.	Performance
2	The system should be available at working hours.	Availability
3	The minimum period of hire is one day, and all hires are made for complete days.	A constraint on the whole system.